Diagnosis: Interface Energy Loss of the Impedance-Overlap Schwarz Coupling on Nonconforming Meshes, and an Energy-Consistent Redesign

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Abstract

A 4×3×3 parametric study of the impedance-overlap Schwarz coupling on the bent-cantilever release problem (integrator pairs II/IE/EI/EE × overlap widths $\{0.0508, 0.0254, 0.0127\}$ m × mesh ratios $\{1{:}1, 3{:}4, 1{:}2\}$, 1 ms at $\Delta t = 10^{-6}$ s) shows the partition-of-unity energy conserved to 0.1% on conforming meshes but losing 28–71% — and in one case *gaining* 12% — on nonconforming ones. Channel-resolved interface-work instrumentation and an impedance-scale sweep identify the cause: the impedance dashpot $\boldsymbol{Z}(\dot{\boldsymbol{u}}_p - \dot{\boldsymbol{u}})$ acts on an interface velocity jump that cannot vanish at Schwarz convergence when the two trace spaces differ, and because the two subdomains transfer partner data through *independent, non-adjoint* interpolations, the dashpot work is not even sign-definite. The optimized-Schwarz, multi-time-step, and discontinuous-Galerkin literatures converge on the same two structural requirements for an energy-consistent coupling: (i) the kinematic transfer and the force transfer between the two sides must be L^2-adjoints through a single variational cross-mass matrix, which makes the interface power telescope to a sign-definite dashpot on the common-space jump plus a *conservative* spring term; and (ii) the dashpot must act only on content both trace spaces can represent, so that it vanishes at Schwarz convergence instead of persistently absorbing unrepresentable fine-scale content. This note records the evidence, the theory, and a phased redesign.

1 Observed behavior

The benchmark is the bent 3D cantilever released from a parabolic initial displacement, split lengthwise into clamped and free subdomains that overlap, coupled by the impedance-overlap transmission condition

$$\boldsymbol{t} \;=\; \boldsymbol{t}_p \;+\; \boldsymbol{Z}\,(\dot{\boldsymbol{u}}_p - \dot{\boldsymbol{u}}) \;+\; \alpha\,(\boldsymbol{u}_p - \boldsymbol{u}), \qquad \boldsymbol{Z} = Z_p\,\boldsymbol{n}\otimes\boldsymbol{n} + Z_s\,(\boldsymbol{I} - \boldsymbol{n}\otimes\boldsymbol{n}), \tag{1}$$

with $Z_p = \rho c_p$, $Z_s = \rho c_s$ of the partner material (Lysmer–Kuhlemeyer split), $\alpha = 0$ throughout this study, and subscript p denoting partner fields evaluated at this subdomain's Schwarz boundary. Energy is the Arlequin partition-of-unity total (overlap counted once), normalized by each run's own $E(0) \approx 1506$ J.

Drift $E(1\,\mathrm{ms})/E(0) - 1$, in percent, pointwise transfer:

overlap	ratio	II	IE	EI	EE
0.0508 (max)	1:1	−0.07	−8.9	−0.2	+8.7
	3:4	+1.1	−8.7	−0.9	−12.0
	1:2	+12.5	−26.5	−53.8	−57.7
0.0254 ($\frac{1}{2}$)	1:1	−0.07	+5.0	−0.4	+10.2
	3:4	−28.4	−6.0	−22.3	−13.9
	1:2	−17.3	−18.5	−58.9	−45.7
0.0127 ($\frac{1}{4}$)	1:1	−0.02	−15.7	−6.1	+3.6
	3:4	−44.5	−40.6	−41.1	−31.4
	1:2	−63.8	−64.3	−70.8	−61.9

Three regimes: (a) conforming (1:1) implicit-free-subdomain runs conserve to $\leq 0.1\%$ at every overlap; (b) nonconforming runs lose tens of percent, worse with mesh ratio and with smaller overlap; (c) a minority of cases, including II at max overlap 1:2, *gain* energy. Variational (`transfer: variational`) rescues the small-overlap losses dramatically ($-44.5 \to -4.4\%$ at $\frac{1}{4}$/3:4) but regresses elsewhere ($+12.5 \to -50.7\%$ at max/1:2): it improved 17 of 24 nonconforming cases, mean |drift| $33.4 \to 25.4\%$ — helpful, not structural.

2 Empirical diagnosis

2.1 Channel-resolved interface work

The instrumentation (`NORMA_IMPEDANCE_WORK_CSV`) decomposes each subdomain's interface power $P = \sum_i \dot{\boldsymbol{u}}_i \cdot \boldsymbol{f}_i^{\mathrm{bc}}$ into the traction, dashpot, and Robin channels of (1). For the exact monodomain solution the dashpot and Robin channels vanish identically, so their work is purely spurious. Integrated over 1 ms (II runs; work summed over both subdomains):

case	ΔE blended	dashpot work	traction work	jump RMS
conforming ($\frac{1}{2}$, 1:1)	−1 J	−1.8 J	+70 J	~ 1
nonconf. $\frac{1}{2}$, 3:4 (pointwise)	−428 J	−586 J	+92 J	~ 20
nonconf. $\frac{1}{4}$, 3:4 (pointwise)	−515 J	−877 J	+341 J	~ 16
nonconf. max, 1:2 (pointwise)	+188 J	**+554** J	−211 J	~ 80
nonconf. $\frac{1}{4}$, 3:4 (variational)	−67 J	−864 J	+780 J	~ 26

Three facts follow. First, on conforming meshes the dashpot channel is negligible (−1.8 J) and the coupling is conservative: the jump $\dot{\boldsymbol{u}}_p - \dot{\boldsymbol{u}}$ closes at Schwarz convergence. Second, on nonconforming meshes the dashpot channel carries the loss: the jump RMS is an order of magnitude larger and never closes — the coarse trace space cannot represent the fine side's boundary velocity, and the dashpot persistently absorbs the difference, exactly as a Lysmer–Kuhlemeyer boundary absorbs outgoing content. Third, and decisively for the mechanism: in the max/1:2 case the "dashpot" does +554 J of *positive* work. A dissipator can never do positive work; this proves the discrete dashpot term is not a quadratic form. Because each subdomain interpolates partner fields with its own, unrelated operator, the two sides' dashpot terms do not pair, and the total is sign-indefinite.

The variational row explains why variational transfer helps only sometimes: its dashpot still drains −864 J, but the consistent (characteristic-transfer) traction returns +780 J. The rescue is an accidental near-cancellation of two large opposing channels, not a removal of either defect.

2.2 Impedance-scale sweep

Scaling both impedances by $s \in \{1, 0.5, 0.25\}$ on the $\frac{1}{4}$/3:4 II case (pointwise):

impedance scale s	1.0	0.5	0.25
drift at 1 ms	−44.5%	−40.5%	−29.2%

The loss decreases monotonically as the dashpot is weakened — confirming the dashpot as the carrier of the loss — but sublinearly: the dissipated power is $Z\,[\![\dot{u}]\!]^2$ and the jump itself grows as the absorber detunes (reflection increasing as $Z \to 0$), so the product falls more slowly than Z. This is the signature of an absorbing-boundary mechanism rather than of a fixed damping force, consistent with the channel decomposition above.

3 What the literature says

Four independent lines of analysis (all in `~/Downloads/Optimized-Schwarz-Methods`) agree on the structure of the problem and of its solution.

3.1 The interface-work pairing identity

For Newmark-family subdomains coupled by interface tractions, the discrete energy balance over a step contains the interface term (Gravouil–Combescure 2001, Eq. (34); Karimi–Nakshatrala 2014, Eq. (84))

$$E_{\text{int}} = \sum_{\text{sides } k} \frac{1}{\Delta t_k} [\![\dot{\boldsymbol{u}}_k]\!]^T \boldsymbol{C}_k^T [\![\Lambda_k]\!], \tag{2}$$

the work of the transmitted tractions through the kinematic jump, each side on its own grid. It vanishes iff the traction is a *single-valued* field acting equal-and-opposite on the two sides through *adjoint* trace operators, with velocity continuity enforced at instants shared by both time grids. Gravouil–Combescure prove their subcycled coupling makes $E_{\text{int}} \le 0$ (dissipative, never a source); Prakash–Hjelmstad restructure the intermediate tractions so it telescopes to exactly zero. Neither treats spatially nonconforming interfaces; both warn that breaking the pairing leaves E_{int} of indefinite sign — precisely the $+554$ J observation.

3.2 Representability: the converged nonconforming solution

Gander–Halpern–Nataf (SINUM 2003) analyze discrete Schwarz waveform relaxation with impedance transmission on nonmatching grids. Their key statements: the choice $Z = \rho c$ of the *partner* side is the exact transparent operator in 1D and optimal-in-class generally (§3, Eq. (3.10)) — so the impedance choice is not the problem; their nonmatching-grid convergence proof requires the intergrid transfer to be an L^2 *contraction* (projection qualifies, Eq. (5.24); pointwise interpolation does not); and, for nonconforming grids, "the coupled system *defines* the solution when converged" (§5.4) — there is no monodomain scheme it equals. Continuity at the converged fixed point holds only *through the projection*: the component of the fine trace orthogonal to the coarse trace space is invisible to the partner and never matched. The impedance dashpot then acts on that persistent component as a genuine absorbing boundary — sign-definite dissipation that is the price of stability, not an iteration defect. The cure they and Achdou–Japhet–Maday–Nataf (Numer. Math. 2002) point to is not a better Z but a *common trace space* for the datum entering the dashpot.

3.3 The DG telescoping condition

Energy-stable discontinuous-Galerkin couplings (Appelö–Wang 2019; Duru et al. 2021; Liu et al. 2023) make the interface energy identity exact by construction: a single "starred"/Riemann state is computed on a common description of the interface (the mesh intersection) and enters *both* sides' weak forms; the interface power then telescopes to

$$P \;=\; -\int_{\Gamma} [\![\dot{\boldsymbol{u}}]\!]^T \boldsymbol{Z} [\![\dot{\boldsymbol{u}}]\!]\,\mathrm{d}S \;\le\; 0 \qquad \text{(or exactly 0 for the central variant)}, \tag{3}$$

with the dissipation confined to the jump and vanishing under refinement and at convergence. The precise discrete requirement: if side k receives partner kinematics through $\Pi_{j\to k}$, the transfers must be L^2-adjoints through the variational cross-mass matrix,

$$\boldsymbol{M}_1 \Pi_{2\to 1} \;=\; \big(\boldsymbol{M}_2\,\Pi_{1\to 2}\big)^T \;=\; \boldsymbol{B}, \qquad B_{mn} = \int_{\Gamma} \varphi^1_m\,\varphi^2_n\,\mathrm{d}S, \tag{4}$$

with $\boldsymbol{M}_k$ the side-k boundary mass matrices and the integrals evaluated on the common refinement. Two independent pointwise interpolations violate (4), the cross terms do not cancel, and the interface term has no sign. Liu et al. additionally warn that nonconforming cross-interpolation matrices are catastrophically ill-conditioned and that interface unknowns may need implicit treatment even between explicit interiors (their IMEX-Newmark) — directly relevant to the conforming IE/EE growth (+5 to +10%), where the explicitly-evaluated dashpot at lagged time levels injects energy that central-difference ($\gamma = \frac{1}{2}$, zero algorithmic damping) never removes.

4 Synthesis: why each observed regime occurs

Conforming, implicit. Transfer is the identity (node-aligned), the adjoint condition (4) holds trivially, the jump closes at Schwarz convergence, the dashpot does no work: conserved.

Nonconforming, dissipative majority. The jump cannot close (representability); the dashpot absorbs the unrepresentable content persistently. Loss grows with mesh ratio (more unrepresentable content) and shrinks with overlap width (the overlap buffers the boundary from the fine content, and a wider overlap gives the coarse subdomain more elements to express the transferred field).

Nonconforming, growth minority. Non-adjoint transfer makes the dashpot term sign-indefinite; with the extreme 1:2 ratio at max overlap the positive branch dominates. Not stochastic — structural.

Conforming, explicit free subdomain (IE/EE). Time-level, not space, pairing failure: the dashpot force is evaluated explicitly at lagged partner velocity; central difference has no algorithmic damping to absorb the error.

Variational rescue. The variational (L^2) projection is a contraction and the characteristic transfer restores much of the traction consistency, but the dashpot still sees the full jump; the outcome rides on near-cancellation of two large channels.

5 Redesign: an energy-consistent impedance overlap

Phase 1 — adjoint-paired variational transfer. Per Schwarz boundary, assemble the variational cross-mass $\boldsymbol{B}$ of (4) on the common refinement of this side's boundary facets against the partner's trace (machinery largely present in `get_overlap_rectangular_projection_matrix`), and route *all* partner quantities — traction, velocity, displacement — through the adjoint pair $\Pi_{2\to 1} = \boldsymbol{M}_1^{-1}\boldsymbol{B}$, $\Pi_{1\to 2} = \boldsymbol{M}_2^{-1}\boldsymbol{B}^T$. Never form nodal interpolation operators (ill-conditioning); solve with the mass matrices. This restores the pairing that makes the traction exchange cancel and the dashpot sign-definite (3), eliminating the growth regime.

Phase 2 — dashpot on the representable jump *(implemented; falsified).* Replace $\boldsymbol{Z}(\dot{\boldsymbol{u}}_p - \dot{\boldsymbol{u}})$ by $\boldsymbol{Z}\,\widehat{\Pi}\,(\dot{\boldsymbol{u}}_p - \dot{\boldsymbol{u}})$, where $\widehat{\Pi}$ is the W-orthogonal projection onto the transferable span — the projector already built by the content-absorption machinery (`compute_impedance_overlap_schwarz_projectors!`); implemented as `representable dashpot: true` on the BC (a strict no-op on node-aligned interfaces). The premise was that the persistent jump is trace content the coarse space cannot represent, so that filtering it would let it reflect (conserving) instead of being absorbed. *Measurement falsifies the premise*: on the $\frac{1}{4}$/3:4 case the fine side's $\widehat{\Pi}$ is a genuine projection ($\|\widehat{\Pi} - I\|/\sqrt{n} = 0.48$) yet the dashpot work is unchanged ($-1066 \to -1065\,\mathrm{J}$) — the jump lies *inside* the representable subspace. The persistent jump is therefore not a spatial-representability defect but a *solution-content* defect: coarse-grid dispersion phase-lags the transiting pulse, so at Schwarz convergence the two discrete solutions disagree by a smooth, in-band field no static spatial filter can remove. Where $\widehat{\Pi}$ does bite (max/1:2), per-side filtering worsens the transfer asymmetry and the drift ($+12.5 \to +45.6\%$). The option is retained (default off) as a diagnostic of *where* the jump lives, and the dimensional caveat stands: the partner's volume-trace span on a boundary with few nodes is generically onto, so trace-representability is the wrong notion of "what the partner can carry" — the right notion is dynamic (dispersion band), not static.

Phase 3 — harmonic-mean impedance. Use $Z = Z_1 Z_2/(Z_1 + Z_2)$ per wave family (the Riemann-solver weighting of Duru et al. and Liu et al.), reducing to $Z/2$ for identical materials and correct for dissimilar ones; the current one-sided partner impedance over- or under-damps whenever materials or resolved impedances differ.

Phase 4 — implicit interface for explicit subdomains. Evaluate the interface (dashpot) terms implicitly at the synchronization instants even when a subdomain's interior is explicit (Liu's IMEX-Newmark; equivalently Karimi–Nakshatrala's velocity continuity at shared instants), targeting the conforming IE/EE growth.

Regression metric. Keep the channel-resolved interface-work CSV as a test: after Phase 1–2, the dashpot work must be ≤ 0 in every configuration (sign test) and $\to 0$ with Schwarz tolerance on conforming meshes; the study's 36-case drift table is the acceptance benchmark.

The alternative structural cure — a single independently discretized frame (variational multiplier) field on each Schwarz boundary with both sides coupling to it adjointly (the localized-Lagrange-multiplier construction of Park–Felippa, applied in Dvorak et al. 2023) — achieves the same pairing with an extra interface field, and is the natural fallback if the in-place adjoint retrofit of Phase 1 proves intrusive.